Nifty

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Document details

Categories of variables	Longitudinal regional amyloid-beta PET [18F]florbetapir SUVRs and amyloid-beta status for phase 1 and 2 Insight 46 scans.
Summary of work undertaken	Cleaning/derivation work undertaken in R (Rmarkdown script provided). First, source data files were combined. Each source file contains regional SUVRs from a different methodology (different image reconstruction, reference region and partial volume correction). For sessions that had no usable raw list-mode PET data, NiftyPET reconstruction with psudo-CT (pCT) was not possible. Therefore, in the 36 instances where there were acceptable console reconstructions but missing NiftyPET pCT SUVRs, we imputed NiftyPET pCT SUVR from the console SUVR using the prediction from linear regression based on the high correlation between console and NiftyPET pCT SUVRs in the whole cohort (R2 between 0.93 and 0.99 for all methods). Linear regression was used as some variables were too highly correlated to use multiple imputation. Following imputation, positive or negative amyloid status was determined using a Gaussian-mixture modelling (see below). Centiloid values were calculated (see below). Final dataset was then ordered and exported in wide format. See below for details of methodology.
Names of source variables	Each, source, file, contains, the, following, variables:, subject, session, session_date, exclude, frontal, parietal, precuneus, occipital, temporal, antmidcingulate, postcingulate, insula, hippocampus, thalamus, caudate, putamen, composite, cortical_gm, subcortical_gm, Separate, file, for, each, PET, reconstruction, reference, region, or, PVC, method
Output variables	See, data, dictionary, for, full, list, of, names, and, descriptions, in, order:, \\icarus\1946\Test_Data_and_Video_Files\Phase_2\Imaging, Data\, Phase2_Data_Freeze\PHASE2_AMYLOID_SUVR_STATUS_20220809-150732\Insight46_phase2_amyloid-PET_NiftyPET_regional_suvr_20220809-150732_DATADICTIONARY.csv
Papers using these variables	N/A

Methods

** Image Acquisition** Scanning was performed on a Biograph mMR 3T PET/MRI scanner (Siemens Healthcare, Erlangen). Amyloid-beta pathology was assessed over a 10-min period, ~50 min after intravenous injection of ~370 MBq [18F]florbetapir (Amyvid). Attenuation maps are computed from the T1-weighted and T2-weighted volumetric scans using a multi-atlas CT synthesis method, also known as pseudo-CT (pCT). PET data, acquired in list-mode, was reconstructed offline using NiftyPET software with ordered subset expectation maximisation algorithm (14 subsets, 4 iterations, 4mm Gaussian kernel) using the pCT for attenuation correction. PET images were also reconstructed on the scanner console using ultra-short echo time (UTE) MR attenuation correction. Although reconstruction with pCT from list-mode has been shown to be more accurate than UTE when compared to gold-standard CT (Burgos et al., 2014), correlations between SUVRs from each reconstruction method are high (R2>0.93) allowing missing list-mode SUVRs to be predicted from console SUVRs using linear regression.

Image Processing

T1-weighted MRI scans were parcellated using geodesic information flows (GIF) v3 (Cardoso et al., 2015) and then parcellations were rigidly registered and resampled with nearest-neighbour interpolation to PET native space using NiftyReg software. Standardised uptake value ratio (SUVR) images were then created by dividing the whole image by mean uptake in a GIF reference region. SUVRs were calculated using four different reference regions: white matter (eroded by 1 voxel in PET space; WM), whole cerebellum (WC), cerebellar grey matter (CGM) and pons. SUVRs were also calculated with partial volume correction (PVC) applied, which aims to use anatomical information from MRI to reduce spill-in/out of PET signal in adjacent regions due to the low spatial resolution of PET. For PVC, PET images were resampled to T1-space and the GIF regions were used with the iterative-Yang method with a 6.8 mm3 kernel optimised for the PET/MR scanner (Hutton et al., 2013). For each method, SUVRs were then extracted from a GIF composite cortical target region closely matched to FreeSurfer regions used in Landau et al. (2013). The composite consists of a volume weighted average of uptake in lateral & medial frontal, anterior & posterior cingulate, lateral parietal, and lateral temporal regions. Separate regional values were also extracted from the precuneus, posterior cingulate, anterior to mid-cingulate, frontal, parietal, temporal, insula and occipital cortex.

Statistical analysis

Amyloid-beta status was then calculated separately for each region and methodology using Gaussian-mixture modelling applied to SUVRs with two distributions, with the cutpoint set to the 99th percentile of the lower Gaussian. Cutpoints and positivity rates for each region and method are provided in a separate spreadsheet (Insight46_phase2_amyloid-PET_NiftyPET_suvr_rates_20220809-150732.csv). Centiloid (CL) values were then calculated to provide a measure comparable between datasets compared to SUVR values (Klunk et al., 2015). CL values were only calculated for whole cerebellum and white matter referenced SUVRs using methodology found in Coath et al. (2022). The adjustment for white matter SUVRs was re-calculated in the R script using this NiftyPET dataset due to the difference in reconstruction from the cross-sectional dataset. See Table for CL equations and cutpoints.

Table. Composite region Centiloid (CL) equations and cutpoints:

Reference region/PVC	SUVR cutpoint	Conversion equation	Centiloid cutpoint
Whole cerebellum	1.071	SUVR*194.0 - 194.7	13.0
Whole cerebellum/PVC	1.032	SUVR*106.9 - 98.4	11.9
White matter	0.605	SUVR*433.2 - 239.4	22.6
White matter/PVC	0.659	SUVR*220.8 - 126.5	18.9

References

Burgos, Ninon, et al. "Attenuation correction synthesis for hybrid PET-MR scanners: application to brain studies." IEEE transactions on medical imaging 33.12 (2014): 2332-2341. Cardoso, M. Jorge, et al. "Geodesic information flows: spatially-variant graphs and their application to segmentation and fusion." IEEE transactions on medical imaging 34.9 (2015): 1976-1988. Coath, William, et al. "Operationalising the Centiloid Scale for [18F] florbetapir PET Studies on PET/MR." medRxiv (2022). Hutton, Brian F., et al. "What approach to brain partial volume correction is best for PET/MRI?." Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment 702 (2013): 29-33. Klunk, William E., et al. "The Centiloid Project: standardizing quantitative amyloid plaque estimation by PET." Alzheimer's & dementia 11.1 (2015): 1-15. Landau, Susan M., et al. "Amyloid-β imaging with Pittsburgh compound B and florbetapir: comparing radiotracers and quantification methods." Journal of Nuclear Medicine 54.1 (2013): 70-77. Markiewicz, Pawel J., et al. "NiftyPET: a high-throughput software platform for high quantitative accuracy and precision PET imaging and analysis." Neuroinformatics 16.1 (2018): 95-115.

Variables in this document

No variables were detected in this document.